Charles Benello

+1(312) 989-9950 | cmbenello@uchicago.edu | linkedin.com/in/charles-benello | cmbenello.github.io

Education

The University of Chicago — Chicago, IL
Master of Science in Financial Mathematics — *Expected Dec 2026*

- Wayne Booth Graduate Student Teaching Prize (2026) - UChicago's top graduate teaching award
- Coursework: ML in Finance, Computing for Finance, Options Pricing, Probability & Stochastic Processes, Reinforcement Learning, Agentic AI

The University of Chicago | GPA: 3.75/4.0 — Chicago, IL
Bachelor of Science in Mathematics and Computer Science (Systems), Minor in Korean — *Jun 2025*

- Coursework: Algorithms, Database Systems, Distributed Systems, Concurrency Control, Operating Systems

Experience

Quantitative Researcher — Oct – Dec 2025
Bank of America (UChicago Project Lab) — *Chicago, IL*

- Built an AI-driven platform to rebalance high-quality liquid asset portfolios under capital, liquidity, and net interest income constraints, generating trade recommendations that minimize risk over 6-month projections.
- Engineered a multi-agent LLM debate system where competing agents generate macro stress scenarios and produce probability-weighted reallocation recommendations.
- Shipped a Next.js scenario explorer, Flask yield curve dashboard, and FastAPI service exposing the optimizer to downstream tools; built a daily UST yield curve ingestion pipeline pulling real-time market, macro, and news data.

Quantitative Researcher — Mar 2026 – Present
PayPal (UChicago Project Lab) — *Chicago, IL*

- Built a self-supervised transaction foundation model (TabBERT-style hierarchical transformer, ∼5M params) on 14.5M unlabeled transactions; achieved AUC 0.974 / F1 0.822 on downstream fraud detection at 0.12% positive rate.
- Ran 87+ ablations on UChicago's Midway3 SLURM cluster across architecture, tokenization, time encoding, and masking; implemented per-field masking and SPRM from PANTHER (NeurIPS 2025) and PRAGMA (Revolut 2026).
- Evaluated learned embeddings against XGBoost and MLP baselines on highly imbalanced data, with focus on probability calibration in the tails.

Software Engineering Intern — Feb 2026 – Present
Alter Domus — *Chicago, IL*

- Lifted document extraction accuracy from 50% to 92–97% by architecting a multi-stage AI pipeline with schema-enforced structured output and cross-reference verification for 100+ page financial filings.
- Built an LLM-as-a-Judge evaluation framework with automated benchmarks for faithfulness, numerical accuracy, and completeness across document types.

Database Research Assistant — Apr 2023 – Present
ChiData Research Lab, University of Chicago — *Chicago, IL*

- First-authored VLDB 2026 paper (CrocSort) on parallel external merge sort in Rust; co-authored CIDR 2025 paper on resource-adaptive query execution; sole-authored HPTS 2026 paper on cost-based sort configuration.

GPU Systems Researcher — Jun – Sep 2025
EPFL (DIAS Lab) — *Lausanne, Switzerland*

- Cut out-of-memory query times by 50–200% by engineering a macroblock-based GPU decompression strategy in CUDA/C++; built cost-based simulation models predicting query performance across workload complexities for adaptive strategy selection.

Projects

Transit Coverage Optimizer | *Python* | cmbenello.github.io/covtsp — 2026

- Beat the Guinness World Record by 60 minutes (16h45m vs. 17h45m) with an automated Covering TSP pipeline: fetches GTFS feeds and Google Maps walking data, builds a time-expanded graph (160K nodes, 1.26M edges), and runs k-NN with urgency scoring and epsilon-greedy search.

MiniMetro RL | *Python, PyTorch* | cmbenello.github.io/MiniMetro — 2025

- Built a custom Gymnasium environment simulating the transit-building game Mini Metro, with a full game engine and trained a PPO agent with action masking and reward shaping to learn network construction strategies.

Technical Skills

Languages: Python, Rust, C/C++, SQL, CUDA, TypeScript
Frameworks & Tools: PyTorch, XGBoost, NumPy/Pandas, scikit-learn, statsmodels, FastAPI, Next.js, Flask, Docker, Git, Linux, PostgreSQL, DuckDB, Databricks
Concepts: Time Series Analysis, Backtesting, Trading Signals, Multi-Agent Systems, LLM Evaluation, Self-Supervised Learning, Reinforcement Learning, Statistical Modeling

Publications

[1] R. Otaki*, **C. Benello***, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026*. (*Equal contribution)
[2] **C. Benello**. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." *HPTS 2026*.
[3] R. Otaki, J.H. Chang, **C. Benello**, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution." *CIDR 2025*.

Additional

Citizenship: US, UK, Italy | **Languages**: Korean, Italian (Intermediate); French (Basic) | **Interests**: Running (Chicago Marathon '22, '23, '24)